

Quiz 4

● Graded

Student

Myles Sartor

Total Points

13.5 / 16 pts

Question 1

Problem 1

3.5 / 4 pts

– 0 pts Correct

– 0.5 pts (a) correct z score for 4 is 1

– 0.5 pts (a) correct z score for 7 is 2

– 0.5 pts (a) $\phi(2) - \phi(1) = (0.95 - 0.68)/2 = 0.27/2 = 0.135$: incorrect phi values

– 0.5 pts (a) $\phi(2) - \phi(1) = (0.95 - 0.68)/2 = 0.27/2 = 0.135$: forgot to divide by 2

– 0.5 pts (b) $|x - 1| > 6$ is $x - 1 > 6$ or $x - 1 < -6$

– 0.5 pts (b) $P(|x - 1| > 6) = P(x - 1 > 6) + P(x - 1 < -6)$

– 0.5 pts (b) $P(|x - 1| > 6) = P(x - 1 > 6) + P(x - 1 < -6) = 1 - \phi(2) + \phi(-2)$: incorrect z scores

✓ – 0.5 pts (b) $P(|x - 1| > 6) = P(x - 1 > 6) + P(x - 1 < -6) = 1 - \phi(2) + \phi(-2) = 1 - 0.95 = 0.05$

– 4 pts wrong process

– 0.5 pts (a) why did you divide by 3?

– 0 pts Click here to replace this description.

Question 2

Problem 2

3 / 4 pts

– 0 pts Correct

– 0.5 pts part a: minor error in expression

– 1 pt part a: major error in expression.

– 0.5 pts part b: minor error in expression.

✓ – 1 pt part b: major error in expression

Question 3

Problem 3

7 / 8 pts

- 0 pts Correct

- 1 pt (a) $\lambda = 1.5$

- 1 pt (a) For $X_1 = \text{\#complaints within 5 days}$, find $P(X_1 = 0)$

- 1 pt (a) answer: $e^{-1.5}$

(b)

- 1 pt (b) For $X_2 = \text{\#complaints within a week}$, $P(X_2 = 0) = e^{-2.1}$

- 1 pt (b) For $Y = \text{\#complaint-free weeks in 10 weeks}$, $Y \sim \text{Bin}(10, e^{-2.1})$

- 1 pt (b) answer for $P(Y=4)$

- 1 pt (c) For $Z = \text{\#complaint-free weeks until fourth complaint-free week}$, $Z \sim \text{NBin}(r=4, p=e^{-2.1})$

✓ - 1 pt (c) answer for $P(Z=10)$

- 8 pts Incorrect

✖ **Problem 1.** If X is a normal random variable with parameters $\mu = 1$ and $\sigma^2 = 9$. Find

(a) $P(4 < X < 7)$

$$N \sim (1, 9) \quad \sigma = 3$$

(b) $P(|X - 1| > 6)$

$$N \sim (\mu, \sigma^2) \quad z = \frac{x - \mu}{\sigma}$$

✖ You should use the 68-95-99.7 rule to simplify your final answers for both parts.

$$z = \frac{x - 1}{3}$$

$$a) P(4 < X < 7) \rightarrow P\left(\frac{4-1}{3} < z < \frac{7-1}{3}\right)$$

$$\phi(0) = \frac{1}{2}$$

$$= P(1 < z < 2)$$

$$\phi(2) - \phi(1)$$

$$= \frac{0.95 - 0.68}{2} = \frac{0.27}{2} = \boxed{0.135}$$

$$b) P(|X - 1| > 6)$$

$$1 - P(-6 < X - 1 < 6)$$

$$= 1 - P(-5 < X < 7)$$

$$= 1 - P\left(\frac{-5+1}{3} < z < \frac{7-1}{3}\right)$$

$$= 1 - P\left(-\frac{4}{3} < z < 2\right)$$

$$= 1 - \phi(2) - \phi\left(-\frac{4}{3}\right)$$

$$= \frac{1 - 0.95 - \phi\left(-\frac{4}{3}\right)}{2}$$

$$\boxed{\frac{0.05 - \phi\left(-\frac{4}{3}\right)}{2}}$$

- Problem 2.** Suppose that the probability that a duck hunter will successfully hit a duck is 0.6 on any given shot. Suppose also that the outcome of each shot is independent from the others.

$$p = 0.6$$

4 trials until a success

trial

(a) What is the probability that the first successful hit would be on the fourth shot?

(b) what is the probability that the third successful hit would be on the ninth shot?

9 trials until another success

$$\binom{4}{1} (0.6)^1 (0.4)^{4-1} = \binom{4}{1} (0.6) (0.4)^3$$

$$\binom{9}{3} (0.6)^3 (0.4)^{9-3} = \binom{9}{3} (0.6)^3 (0.4)^6$$

$$c) P(X=k) = p(1-p)^{k-1} \rightarrow 0.6 (1-0.4)^{4-1}$$

$$P(X=4) = (0.6)(0.4)^3$$

$$b) P(X=k) = 0.6 (1-0.4)^{9-1}$$

$$P(X=9) = (0.6)(0.4)^8$$

Problem 3. Customer complaints to a tech support center follow a Poisson distribution with an average rate of 2.1 complaints per week.

$X \sim \text{Pois}(\lambda)$ (a) What is the probability that no complaints will be received for the next 5 days?

$X \sim \text{Bin}(n, p)$ (b) What is the probability that there will be exactly 4 complaint-free weeks during the next ten-weeks
4 successes in 10 trials

$X \sim \text{NBin}(r, p)$ (c) Assume a person just started to work here. What is the probability that the week 10 is the fourth complaint free week?

$$P(X=k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

$$a) P(X=0) = \frac{e^{-1.5} 1.5^0}{0!} = e^{-1.5}$$

$$\lambda = 2.1$$

$$\lambda_{\text{per day}} = 0.3$$

$$\frac{2.1}{7} = 0.3$$

$$0.3 \cdot 5 = 1.5 \Rightarrow 5 \text{ days}$$

$$b) P(X=0) = \frac{e^{-2.1} 2.1^0}{0!} = e^{-2.1}$$

$$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k} \rightarrow P(X=4) = \binom{10}{4} (e^{-2.1})^4 (1 - e^{-2.1})^{10-4}$$

$$= \binom{10}{4} (e^{-2.1})^4 (1 - e^{-2.1})^6$$

$$c) P(X=10) = \binom{10-1}{4-1} (e^{-2.1})^3 (1 - e^{-2.1})^{9-3}$$

$$= \binom{9}{3} (e^{-2.1})^3 (1 - e^{-2.1})^6$$

$$P(X=k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}$$

Quiz 4

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Table 1: Discrete Distributions

Distribution	Known Abbreviation	Pmf	Expected Value	Variance
Bernoulli	$X \sim \text{Bern}(p)$	$p_X(k) = p^k(1-p)^{1-k}$ where $k = 0, 1$	$\mu = p$	$\sigma^2 = p(1-p)$
Binomial	$X \sim \text{Bin}(n, p)$	$p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}$ where $k = 0, 1, \dots, n$	$\mu = np$	$\sigma^2 = np(1-p)$
Geometric	$X \sim \text{Geom}(p)$	$p_X(k) = p(1-p)^{k-1}$ where $k = 1, 2, \dots$	$\mu = \frac{1}{p}$	$\sigma^2 = \frac{1-p}{p^2}$
Negative Binomial	$X \sim \text{NBin}(r, p)$	$p_X(k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}$ where $k = r, r+1, \dots$	$\mu = \frac{r}{p}$	$\sigma^2 = \frac{r(1-p)}{p^2}$
Poisson	$X \sim \text{Pois}(\lambda)$	$p_X(k) = e^{-\lambda} \frac{\lambda^k}{k!}$ where $k = 0, 1, 2, \dots$	$\mu = \lambda$	$\sigma^2 = \lambda$

Table 2: Continuous Distributions

Distribution	Known Abbreviation	Pdf	Expected Value	Variance
Normal distribution	$X \sim N(\mu, \sigma^2)$	$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}; -\infty < x < \infty$	μ	σ^2
Exponential distribution	$X \sim \text{Exp}(\lambda)$	$f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$

